

Kai-Lin Chen

<https://hiimtiger.github.io/>

Github

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Education	Cornell University, Ithaca, New York, United States 2026 - 2027 (Expected) Master of Engineering, Department of Electrical and Computer Engineering National Taiwan University, Taipei, Taiwan September 2022 - 2026 Bachelor of Science and Engineering, Department of Biomedical Engineering GPA: 3.83/4.30 (Updated 2025 Spring) Selected Courses: Programming Languages 2023 Fall(A+), Digital Engineering Signal and Systems 2024 Spring(A+), Fundamentals of Biomedical Image Processing 2024 Fall(A+), Machine Learning 2024 Fall(A-), Medical Image Analysis 2025 Spring(A+), Introduction to Biophotonics 2025 Spring(A+), Data Structure 2025 Spring(A-)
Technical Skills	• Programming Languages: Python, MatLab, C++, L^AT_EX, HTML. • Toolboxes: Pytorch, OpenCV, Tensorflow, FreeSurfer.
Research Interests	• Medical Image Analysis: Medical Image Segmentation, Computer Vision, Deep Learning. • Image Optimization: Image Reconstruction, Modality Fusion. • Low-resource Training Methods: Data Augmentation, Pseudo-dataset Simulation
Publications	Kai-Lin Chen, Yu-Nong Lin, Pen-hsiu Grace Chao, Kevin T. Chen <i>A Low Resource Training Strategy for Cell Segmentation Using Patch-Based Attention U-Net.</i> MIDL: Medical Imaging with Deep Learning. [Short Paper] (2025).
Research Experience	NTU Multimodal Medical Imaging Optimization Lab with Dr. Kevin T. Chen 2023-Present Undergraduate Research Assistant Worked on <i>Segmenation toolbox development</i> and <i>Full-count PET image reconstruction Projects</i> • Labeled airways in dynamic 3D MRI to model breathing cycles for healthy and OSA patients. • Developed a MATLAB pipeline to segment and measure actin fibers in phase-contrast microscopic images. • Developed data augmentation workflows for low-resource segmentation model training. • Designed a Dynamic Loss function Selection Method to tackle severe class imbalance. • Designed a low-resource segmentation model training pipeline for in-house segmentation model training. (Capable of training with CPU only, workable under data scarcity.) • Developed a low-resource segmentation model training method & packaged as a segmentation model training/inference toolbox PAULow. (Currently deployed in NTU Cell Tissue Engineering Lab) • Constructed low-count Positron Emission Tomography simulation pipeline to generate pseudo-datasets for full-count image reconstruction tasks. Research Center for Applied Sciences at Academia Sinica with Dr. Peilin Chen 07-08/2025 Summer Research Intern Worked on <i>Adenocarcinoma Cells classification Project</i> and <i>Modality Enhancement Project</i>. • Assisted in batch processing of force-sensing microchips as substrates for cell culturing. • Researched on behaviors of HCC827 and A549 carcinoma cell lines. • Proposed a new medical image modality by modality fusion for carcinoma cell classification, fusing cell boundaries and internal force gradients for downstream tasks. • Constructed a CNN classification model for automated carcinoma cell image classification. • Proposed modality enhanced CNN classification accuracy by 37.67%, Modality Fusion Network reduces data processing time by 96%. • Received the silver medal in RCAS Summer Internship Seminar (Poster + Flash talk)

Selected Projects	PAULow: Patch-based Attention U-Net for Low-resource learning 2023 - 2026 - Designed a patch-based preprocessing pipeline and custom Attention U-Net architecture for medical image segmentation. - Proposed a Dynamic Loss Function selection method for severe class imbalance during training. - Trainable with CPU only, with no reliance on pretrained weights; adaptable to diverse ROIs. - PAULow achieved up to 35.62% dice score improvements compared with Cellpose and Stardist. - Enabled finetuning PAULow model and Segment Anything Model in toolbox for stronger performances. Github Link: https://github.com/Hiimtiger/PAULow
	FICM: Force Informed Cell Map Image Synthesis using Attention-gated Dual Encoder U-Net for carcinoma cell line classification 07-08/2025 - Designed a deep learning-based modality fusion framework to generate a new medical image modality for cell mechanobiology. - Achieved 37% improvement in CNN classification accuracy, reduced 96% data processing time. - Concluded that internal force is able to provide features to distinguish carcinoma cell lines. Github Link: https://github.com/Hiimtiger/FICM
	Unsupervised Segmentation of Pathology Images using Haralick Features 2024 Fall - Partnered with graduate students and worked on a research project that uses haralick features for unsupervised segmentation in pathology images. - Applied Haralick features to segment and cluster pathology images without labeled data. - Enabled faster diagnostic workflows through unsupervised learning techniques. - Can adaptively infer the number of clusters for multi-class segmentation tasks, demonstrating potential for scalable segmentation for labeling in clinical settings.
	Brain Tumor Segmentation from MRI using Patch-Based Attention U-Net for Data-Efficient Learning 2025 Spring - adapted PAULow's low-resource and data-efficient pipeline to the BraTS dataset. - Showed strong segmentation performance under limited data conditions (30 v.s. 300 cases). - Achieved higher performance than other U-Net structures in normal & data-limited conditions.
	Blood Clot Formation Modeling with LAMMPs: Thrombosis v.s. Hydrodynamic stress in molecular simulations. 2025 Fall - Visualization of a two-dimensional particle-based simulation of blood clot formation under various flow speed and cohesion strength, rendered using VMD - Highlights how clot growth and mechanical stabilization emerge from the interplay between hydrodynamic transport, adhesion strength, and multistage platelet activation. - Analyzed Stress Evolution and clot size under different settings.
Teaching and Activities	NTUBME Advanced Engineering Mathematics DBME2004, <i>Teaching Assistant</i> 09/2025 - 12/2025 With Prof. Chung-Ming Chen and Prof. Li-An Chu.
	EngLite Event Organizer and Speaker at NTUBME, <i>Event Organizer and Speaker</i> 2024 - 2026 With Prof. Pen-hsiu Grace Chao, hosted multiple English Speaking events @ NTUBME.
	Welcome Week Ambassador & International Student Volunteer at NTU 2022 - 2023 Designed an unofficial most optimal touring route using the Dijkstra's algorithm in National Taiwan University and assisted multiple International students survived through registration, navigating the NTU urban-life circle, and course selection throughout their exchange.
Awards and Fellowships	2025 National Science and Technology Council (NSTC), Undergrad Research Fellowship 2025 RCAS Summer Intern Seminar, Silver Medal in RCAS Summer Internship Seminar (Poster + Flash talk) 2026 NTUBME Undergraduate Capstone Seminar, second place